Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $x_1, x_2, x_3, x_4, x_5, \ldots$ is a geometric progression of natural numbers and $x_1x_2x_3x_4 = 64$ then complete set of solutions of x_4 is

(A) $\{1\}$

(B) $\{8\}$

(C) $\{1, 8\}$

(D) can not be determined

Q.2

If $\frac{a+bx}{a-bx} = \frac{b+cx}{b-cx} = \frac{c+dx}{c-dx} (x \neq 0)$, then a, b, c, d are in

(A) A.P.

(B) G.P.

(C) H.P.

(D) none of these

Q.3

For a positive integer n, let $a(n) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{(2^n)-1}$. Then

(A) $a(100) < 100$

(B) $a(100) > 100$

(C) $a(200) < 100$

(D) none of these

Q.4

If a, b, c are in A.P., then $\frac{1}{\sqrt{a}+\sqrt{b}}, \frac{1}{\sqrt{a}+\sqrt{c}}, \frac{1}{\sqrt{b}+\sqrt{c}}$ are in

(A) A.P.

(B) G.P.

(C) H.P.

(D) none of these

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Q.5

The series $\frac{2x}{x+3}+\left(\frac{2x}{x+3}\right)^2+\left(\frac{2x}{x+3}\right)^3+\ldots\ldots$ will have a definite sum when

(A) $-1<x<3$

(B) $0<x<1$

(C) $x=0$

(D) none of these

Q.6

Two consecutive numbers from $1,2,3,\ldots\ldots.50$ are removed. A.M. of the remaining $\frac{105}{4}$. Then the numbers are numbers is

(A) 7,8

(B) 8,9

(C) 6,7

(D) none of these

Q.7

If $\frac{48}{(2)(3)}+\frac{47}{(3)(4)}+\frac{46}{(4)(5)}+\ldots+\frac{2}{(48)(49)}+\frac{1}{(49)(50)}=\frac{51}{2}+k\left(1+\frac{1}{2}+\frac{1}{3}+\ldots+\frac{1}{50}\right)$ then k is equal to

(A) 2

(B) -1

(C) $-1/2$

(D) 1

Q.8

If $4a^2+9b^2+16c^2=2(3ab+6bc+4ca)$, where a,b,c are nonzero numbers, then a,b,c are in

(A) A.P.

(B) G.P.

(C) H.P.

(D) none of these

Q.9

a,b,c are distinct positive real numbers such that $a>b>c$. If $2\log(a-c),\log\left(a^2-c^2\right),$ $\log\left(a^2+2b^2+c^2\right)$ are in A.P., then

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(A) a, b, c are in A.P.

(B) $\sqrt{a}, \sqrt{b}, \sqrt{c}$ are in A.P.

(C) a, b, c are in G.P.

(D) a, b, c are in H.P.

Q.10

If $a > 0, b > 0, c > 0$ and the minimum value of $a\left(b^2 + c^2\right) + b\left(c^2 + a^2\right) + c\left(a^2 + b^2\right)$ is λabc then λ is

(A) 2

(B) 1

(C) 6

(D) 3

Q.11

In the sequence $1, 2, 2, 4, 4, 4, 4, 8, 8, 8, 8, 8, 8, 8, 8, \ldots\ldots$, where n consecutive terms have the value n, the 1025th term is

(A) 2^9

(B) 2^{10}

(C) 2^{11}

(D) 2^8

Q.12

If n arithmetic means $a_1, a_2, \ldots\ldots, a_n$ are inserted between 50 and 200 and n harmonic means $h_1, h_2, \ldots\ldots, h_n$ are inserted between the same two numbers, then $a_2 h_{n-1}$ is equal to

(A) 500

(B) $\frac{10000}{n}$

(C) 10000

(D) $\frac{250}{n}$

Q.13

If a, b, c are in H.P., then

(A) $\frac{a}{b+c-a}, \frac{b}{c+a-b}, \frac{c}{a+b-c}$ are in H.P.

(B) $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in H.P.

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(C) $a-\frac{b}{2}, \frac{b}{2}, c-\frac{b}{2}$ are in G.P.

(D) All of these

Q.14

If x, y, z are positive numbers in A.P., then consider the following statements:
(A) $\frac{x+y}{(2y-x)}+\frac{y+z}{(2y-z)} \geq 4$
(B) $y \geq \sqrt{xz}$
(C) $\frac{x+y}{(2y-x)}+\frac{y+z}{(2y-z)} \geq 2$
(D) $y^2 \geq \sqrt{xz}$
Then which of the following is always true?

(A) Only A

(B) A and B

(C) A, B and C

(D) $\mathrm{A}, \mathrm{B}, \mathrm{C}$ and D

Q.15

The sum to n terms of the series $\frac{1}{1^3}+\frac{1+2}{1+2^3}+\frac{1+2+3}{1+2^3+3^3}+\ldots \ldots$ is

(A) $\frac{n}{n+1}$

(B) $\frac{n}{2(n+1)}$

(C) $\frac{2n}{n+1}$

(D) $\frac{2}{n(n+1)}$

Q.16

If $[x]$ denotes the greatest integer $\leq x$, then value of $S=[\sqrt{1}]+[\sqrt{2}]+[\sqrt{3}]+\ldots+[\sqrt{2024}]$ is

(A) 59000

(B) 58750

(C) 59730

(D) 65138

Q.17

If $a_1, a_2, \ldots \ldots, a_n$ are in H.P., then the expression $a_1 a_2+a_2 a_3+\ldots \ldots+a_{n-1} a_n$ is equal to

(A) $(n-1)\left(a_1-a_n\right)$

(B) $n a_1 a_n$

(C) $(n-1) a_1 a_n$

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(D) $n\left(a_1 - a_n\right)$

Q.18

A boy goes to school from his home at a speed of $x\ \mathrm{km}/$ hour and comes back at a speed of $y\ \mathrm{km}/$ hour, then the average speed is equal to

(A) A.M. of x, y

(B) G.M. of x, y

(C) H.M. of x, y

(D) none of these

Q.19

Given two numbers a and b. Let A denote that single A.M. and S denote the sum of n A.M.s between a and b, then $\frac{S}{A}$ depends on

(A) n, a, b

(B) n, b

(C) n, a

(D) n

Q.20

If $\tan m\theta = \tan n\theta$, then the general value of θ will be in

(A) A.P.

(B) G.P.

(C) H.P.

(D) none of these

Q.21

If $\log_2(a+b) + \log_2(c+d) \geq 4$. The minimum attainable value of the expression $a+b+c+d$ is

Q.22

The sum of the series $\frac{3}{4} + \frac{5}{36} + \frac{7}{144} + \ldots$ upto n terms is $\frac{n(n+k)}{(n+1)^2}$ then $k =$

Q.23

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The angles of a triangle are in A.P. and the radian measure of the smallest to the degree measure of the mean is $\pi : 200$. The greatest angle in radians is $\frac{p\pi}{q}$ where p and q are coprime, then $p + q =$

Q.24

Let $(a_1, a_2, a_3, \ldots\ldots)$ be a sequence such that $a_1 = 2$ and $a_n - a_{n-1} = 2n$ for all $n \geq 2$. Then $\sum_{i=1}^{20} a_i$ is

Q.25

The sum of the series $\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \ldots.. + \frac{1}{\sqrt{n^2-1}+\sqrt{n^2}}$ upto 300 terms is _____.

Q.26

Let T_r be the r th term of an arithmetic progression, whose first term is a and common difference is d. If for some positive integers $m, n, m \neq n, T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$, then $a - d$ equals

Q.27

If $\sum_{r=1}^{n} t_r = \frac{1}{12}n(n+1)(n+2)$, then the value $\sum_{r=1}^{n} \frac{1}{t_r}$ rounded off to the nearest integer for n = 2021 is _____.

Q.28

If sum of n terms of an A.P. is $3n^2 + 5n$, while its m th term $T_m = 164$, then m is

Q.29

$2^{\frac{1}{4}} \cdot 4^{\frac{1}{8}} \cdot 8^{\frac{1}{16}} \cdot 16^{\frac{1}{32}} \ldots\cdot\cdot\cdot\cdot\cdot$ is equal to

Q.30

A G.P. consists of an even number of terms. If the sum of all the terms is 5 times the sum of the terms occupying odd places, then the common ratio will be equal to

Answers & Solutions

Q.1 Answer:

$\{8\}$

Solution:

Let $x_1 = \frac{a}{r^3} \quad x_2 = \frac{a}{r}, x_3 = ar$ **and** $x_4 = ar^3$

$a^4 = 64 \quad \Rightarrow \quad a = 2\sqrt{2}$

As $x_1, x_2, \cdots$ **is natural number**

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$\frac{2\sqrt{2}}{r^3} = x_1 \quad \Rightarrow \quad r = \sqrt{2}$

For $r = \frac{1}{\sqrt{2}}$, x_5 is not a natural number. So, $x_4 = 2\sqrt{2}(\sqrt{2})^3 = 8$

Q.2 Answer:

G.P.

Solution:

$\Rightarrow \frac{a+bx}{a-bx} = \frac{b+cx}{b-cx} = \frac{c+dx}{c-dx}$

By componendo and dividendo, we get

$\frac{a}{b} = \frac{b}{c} = \frac{c}{d} \Rightarrow \frac{b}{a} = \frac{c}{b} = \frac{d}{c} \Rightarrow a, b, c, d$ **are in G.P.**

Q.3 Answer:

$a(100) < 100$

Solution:

We have

$$a(n) = 1 + \left(\frac{1}{2} + \frac{1}{3}\right) + \left(\frac{1}{4} + \cdots + \frac{1}{7}\right) + \left(\frac{1}{8} + \cdots + \frac{1}{15}\right) + \cdots + \left(\frac{1}{2^{n-1}} + \frac{1}{2^{n-1}+1} + \cdots + \frac{1}{(2^n)-1}\right)$$

$1 + \frac{2}{2} + \frac{4}{4} + \frac{8}{8} + \cdots + \frac{2^{n-1}}{2^{n-1}} = n$

Thus, $a(100) < 100$.

Q.4 Answer:

A.P.

Solution:

$\frac{1}{\sqrt{a}+\sqrt{b}} = \frac{\sqrt{b}-\sqrt{a}}{b-a} = \frac{\sqrt{b}-\sqrt{a}}{d}$ **where d is the c.d of A.P. a, b, c**

$\frac{1}{\sqrt{a}+\sqrt{c}} = \frac{\sqrt{c}-\sqrt{a}}{c-a} = \frac{\sqrt{c}-\sqrt{a}}{2d}$

$\frac{1}{\sqrt{b}+\sqrt{c}} = \frac{\sqrt{c}-\sqrt{b}}{c-b} = \frac{\sqrt{c}-\sqrt{b}}{d}$

Clearly,

$\frac{\sqrt{b}-\sqrt{a}}{d} + \frac{\sqrt{c}+\sqrt{b}}{d} = \frac{\sqrt{c}-\sqrt{a}}{d} = 2\left(\frac{\sqrt{c}-\sqrt{a}}{2d}\right)$

Q.5 Answer:

$-1 < x < 3$

Solution:

Common ratio of given G.P. $= \frac{2x}{x+3}$

For definite sum of infinite G.P., $-1 < \frac{2x}{x+3} < 1$

$\Rightarrow \quad \frac{2x}{x+3} + 1 > 0$ **and** $\frac{2x}{x+3} - 1 < 0 \quad \Rightarrow \frac{3(x+1)}{x+3} > 0$ **and** $\frac{x-3}{x+3} < 0$

$\Rightarrow \quad (x < -3$ **or** $x > -1)$ **and** $(-3 < x < 3) \Rightarrow -1 < x < 3$

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Q.6 Answer:

7,8

Solution:

Let $\mathrm{K}, \mathrm{K}+1$ **from** $1,2,3, \ldots \ldots,, 50$ **be removed**

A.M. of the remaining 48 terms $=\frac{105}{4}$

$\therefore$ **Sum of all the numbers** $=\mathrm{K}+\mathrm{K}+1+48\left(\frac{105}{4}\right)$

$=2 \mathrm{K}+1+12(015)=2 \mathrm{K}+1261$

Also, $1+2+3+\ldots \ldots+50=\frac{50}{2}(1+50)=25 \times 51=1275$

$\therefore \quad 2 \mathrm{K}+1261=1275 \Rightarrow 2 \mathrm{K}=14 \quad \therefore \mathrm{K}=7$

$\therefore$ **Numbers are 7,8**

Q.7 Answer:

-1

Solution:

Let $t_r=\frac{50-r}{r(r+1)}, r \geq 2.$

We have $\sum_{r=2}^{49} \frac{50-r}{r(r+1)}$

$=50 \sum_{r=2}^{49}\left(\frac{1}{r}-\frac{1}{r+1}\right)-\sum_{r=2}^{49} \frac{1}{r+1}$

$=50\left(\frac{1}{2}-\frac{1}{50}\right)-\sum_{r=2}^{49} \frac{1}{r+1}$

$=24-\sum_{r=1}^{50} \frac{1}{r}+\left(1+\frac{1}{2}\right)$

$=\frac{51}{2}-\sum_{r=1}^{50} \frac{1}{r}$

$\therefore k=-1$

Q.8 Answer:

H.P.

Solution:

$(2a)^2+(3b)^2+(4c)^2-(2a)(3b)-(3b)(4c)-(4c)(2a)=0$

$\Rightarrow \quad (2a-3b)^2+(3b-4c)^2+(4c-2a)^2=0$

$\Rightarrow \quad 2a=3b=4c \Rightarrow \frac{a}{1/2}=\frac{b}{1/3}=\frac{c}{1/4}$

Q.9 Answer:

a, b, c are in G.P.

Solution:

$2 \log(a-c), \log\left(a^2-c^2\right), \log\left(a^2+2b^2+c^2\right)$ **are in A.P.**

$\Rightarrow \quad 2 \log\left(a^2-c^2\right)=2 \log(a-c)+\log\left(a^2+2b^2+c^2\right)$

$\Rightarrow \quad 2 \log\left(a^2-c^2\right)-2 \log(a-c)=\log\left(a^2+2b^2+c^2\right)$

$\Rightarrow \quad 2 \log\left(\frac{a^2-c^2}{a-c}\right)=\log\left(a^2+2b^2+c^2\right)$

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$\Rightarrow \quad \log(a+c)^2 = \log\left(a^2+2b^2+c^2\right)$

$\Rightarrow \quad (a+c)^2 = a^2+2b^2+c^2 \Rightarrow ac = b^2 \Rightarrow a, b, c$ **are in G.P.**

Q.10 Answer:

6

Solution:

As A.M. $\geq$ **G.M.**

$\frac{\frac{b}{c}+\frac{c}{b}+\frac{c}{a}+\frac{a}{c}+\frac{a}{b}+\frac{b}{a}}{6} \geq \left(\frac{c}{b}\times\frac{b}{c}\times\frac{c}{a}\times\frac{a}{c}\times\frac{a}{b}\times\frac{b}{a}\right)\frac{1}{6}$

or, $\left(\frac{b^2+c^2}{bc}\right)+\frac{c^2+a^2}{ca}+\frac{a^2+b^2}{ab} \geq 6$

or, $a\left(b^2+c^2\right)+b\left(c^2+a^2\right)+c\left(a^2+b^2\right) \geq 6abc$

$\therefore$ **minimum value of** $a\left(b^2+c^2\right)+b\left(c^2+a^2\right)+c\left(a^2+b^2\right)$ **is** $6abc$

according to question $\lambda abc = 6abc$

$\therefore \lambda = 6$

Q.11 Answer:

2^{10}

Solution:

Let the 1025th **term** $= 2^n$**. Then**

$1+2+4+8+\ldots\ldots+2^{n-1} < 1025 \leq 1+2+4+8+\ldots\ldots+2^n$

$\therefore \quad 2^n-1 < 1025 < 2^{n+1}-1$ or $2^n < 1026 < 2^{n+1}$

$\Rightarrow \quad n = 10$

Q.12 Answer:

10000

Solution:

$a_2 = 50+2\frac{200-50}{n+1} = \frac{50(n+7)}{(n+1)}$

$h_{n-1} = \frac{1}{\frac{1}{50}+\frac{n-1}{n+1}\left(\frac{1}{200}-\frac{1}{50}\right)} = \frac{200(n+1)}{n+7}$

$\Rightarrow \quad a_2 \cdot h_{n-1} = 10000$

Q.13 Answer:

All of these

Solution:

a, b, c **are in H.P.**

$\Rightarrow \quad \frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ **are in A.P.**

$\Rightarrow \frac{a+b+c}{a}, \frac{a+b+c}{b}, \frac{a+b+c}{c}$ **are in A.P.**

$\Rightarrow \frac{b+c}{a}, \frac{c+a}{b}, \frac{a+b}{c}$ **are in A.P.**

$\Rightarrow \quad \frac{b+c}{a}-1, \frac{c+a}{b}-1, \frac{a+b}{c}-1$

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$\Rightarrow$ are in A.P.

$\Rightarrow \quad \frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ are in A.P.

$\Rightarrow$ thus $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in H.P.

and $\frac{a}{b+c-a}, \frac{b}{c+a-b}, \frac{c}{a+b-c}$ are also in H.P.

Lastly, $\left(a-\frac{b}{2}\right)\left(c-\frac{b}{2}\right)=ac-\frac{b}{2}(a+c)+\frac{b^2}{4}$

$=ac-\frac{b}{2}\cdot\frac{2ac}{b}+\frac{b^2}{4}=\frac{b^2}{4}$

$\therefore a-\frac{b}{2}, \frac{b}{2}, c-\frac{b}{2}$ are in G.P.

Q.14 Answer:

A and B

Solution:

A.M. of $x, z=y$, G.M. of $x, z=\sqrt{xz}$ and A.M. $\geq$ G.M.

So $y^2 \geq xz$

and using $(A.M. \geq H.M.), \frac{x+y}{2y-z}+\frac{y+z}{2y-x}=\frac{x+y}{x}+\frac{y+z}{z}=2+y\left(\frac{1}{x}+\frac{1}{z}\right) \geq 4$

and 4 is obviously greater than 2.D is wrong since it fails for the set $(\mathrm{x}, \mathrm{y}, \mathrm{z})=(0.1, 0.2, 0.3)$

Q.15 Answer:

$\frac{2n}{n+1}$

Solution:

$t_n=\frac{1+2+3+\ldots\ldots+n}{1^3+2^3+3^3+\ldots\ldots+n^3}=\frac{\frac{1}{2}n(n+1)}{\frac{1}{4}n^2(n+1)^2}$

$=\frac{2}{n(n+1)}=2\left[\frac{1}{n}-\frac{1}{n+1}\right]$

$$\sum_{k=1}^{n} t_k = 2\left\{\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\ldots\ldots+\left(\frac{1}{n}-\frac{1}{n+1}\right)\right\}$$

$$=2\left(1-\frac{1}{n+1}\right)=\frac{2n}{n+1}$$

Q.16 Answer:

59730

Solution:

For any positive integer k and x, we have

$$[\sqrt{x}]=k \Leftrightarrow k^2 \leq x<(k+1)^2 \Leftrightarrow x \in \left[k^2, k^2+2k\right)$$

$\therefore [\sqrt{x}]=1 \quad$ for $x=1,2,3,$

$$\begin{bmatrix} [\sqrt{x}] & =2 & \text{for } x=4,5,6,7,8 \\ [\sqrt{x}] & =3 & \text{for } x=9,10,11,12,13,14,15, \\ & \vdots & \\ [\sqrt{x}] & =44 & \text{for } x=1936,\ldots,2024 \end{bmatrix}$$

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Thus, $S = (1)(3) + (2)(5) + 3(7) + \ldots + 44(89)$
$= \sum_{k=1}^{44}(k)(2k+1) = 2\sum_{k=1}^{44}(k)^2 + \sum_{k=1}^{44}(k)$
$= \frac{2}{6}(44)(45)(89) + \frac{1}{2}(44)(45)$
$= 59730$

Q.17 Answer:

$(n-1)a_1a_n$

Solution:

$\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \ldots\ldots, \frac{1}{a_n}$ **are in A.P., with common difference** d **(say)**
$\Rightarrow \quad \frac{1}{a_2} - \frac{1}{a_1} = d \Rightarrow a_1 - a_2 = a_1a_2d$
similarly,
$a_2 - a_3 = a_2a_3d$
.....
$a_{n-1} - a_n = a_{n-1}a_nd$
adding all these, we have, $a_1 - a_n = d\{a_1a_2 + a_2a_3 + \ldots\ldots + a_{n-1}a_n\}$ **...(i)**
Also, $\frac{1}{a_n} = \frac{1}{a_1} + (n-1)d \quad d = \frac{a_1 - a_n}{a_1a_n(n-1)}$.
Using in (i) $\Rightarrow \quad a_1a_2 + a_2a_3 + \ldots\ldots + a_{n-1}a_n = a_1a_n(n-1)$

Q.18 Answer:

H.M. of x, y

Solution:

Let the distance of school from home $= d$
and time taken are $t_1 \& t_2$
$t_1 = \frac{d}{x}$ **and** $t_2 = \frac{d}{y}$
Average velocity $= \frac{\text{Total distance}}{\text{total time}} = \frac{2xy}{x+y}$

Q.19 Answer:

n

Solution:

$\because$ A **is the single A.M. between** a **and** b
$\therefore A = \frac{a+b}{2}$
Let $A_1, A_2, \ldots\ldots A_n$ **be** n **A.M.s between** a **and** b
$\therefore a, A_1, A_2, \ldots\ldots A_n, b$ **is an A.P. with common difference** $d = \frac{b-a}{n+1}$

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Now, $S = A_1 + A_2 + \ldots\ldots + A_n = \frac{n}{2}(A_1 + A_n)$

$$n(a + d + a + nd)/2$$

$$= n(2a + (n+1)d)/2$$

$$= n(2a + b - a)/2 \quad [d = (b-a)/(n+1)] = \frac{n}{2}(a+b)$$

$$\therefore S = n\left(\frac{a+b}{2}\right) = nA \Rightarrow \frac{S}{A} = n$$

Q.20 Answer:

A.P.

Solution:

$\tan m\theta = \tan n\theta$

$\Rightarrow \quad m\theta = p\pi + n\theta$

$= \theta = \frac{p\pi}{m-n}$

Hence different values of θ **are in A.P. with** $\frac{\pi}{m-n}$ **as a common difference.**

Q.21 Answer:

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Solution:

$\log_2(a+b)(c+d) \geq 4$

$\Rightarrow \quad (a+b)(c+d) \geq 16$

Now, A.M. $\geq$ G.M.

$\Rightarrow \quad \frac{(a+b)+(c+d)}{2} \geq \sqrt{(a+b)(c+d)}$

$\Rightarrow \quad a+b+c+d \geq 8$

Q.22 Answer:

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Solution:

The given series is $\frac{3}{(1\times2)^2} + \frac{5}{(2\times3)^2} + \frac{7}{(3\times4)^2} + \ldots$

$= \frac{3+(n-1)\times2}{\{n(n+1)\}^2} = \frac{2n+1}{n^2(n+1)^2} = \frac{(n+1)^2-n^2}{n^2(n+1)^2} = \frac{1}{n^2} - \frac{1}{(n+1)^2}$

Its n**th terms** $= \frac{3+(n-1)\times2}{\{n(n+1)\}^2} = \frac{2n+1}{n^2(n+1)^2} = \frac{(n+1)^2-n^2}{n^2(n+1)^2} = \frac{1}{n^2} - \frac{1}{(n+1)^2}$

$\therefore$ **sum of the series** $= \left(\frac{1}{1^2} - \frac{1}{2^2}\right) + \left(\frac{1}{2^2} - \frac{1}{3^2}\right) + \ldots.. + \left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right) = \frac{n(n+2)}{(n+1)^2}$

Q.23 Answer:

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Solution:

Let the angles in degrees in A.P. be taken as $\alpha - \beta, \alpha, \alpha + \beta$

$\therefore \quad A = \alpha - \beta, B = \alpha, C = \alpha + \beta$

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$\therefore \quad A+B+C=180^\circ \Rightarrow \alpha-\beta+\alpha+\alpha+\beta=180^\circ \Rightarrow 3\alpha=180^\circ \Rightarrow \alpha=60^\circ$

Now, $\frac{\pi}{200}=\frac{A \text{ in radians}}{B \text{ in degrees}}=\frac{(60-\beta)\frac{\pi}{180}}{60} \Rightarrow \beta=6^\circ$

$\therefore$ **Greatest angle** $C=\alpha+\beta=60^\circ+6^\circ=66^\circ$

$=66\times\frac{\pi}{180}$ **radian** $=\frac{11}{30}\pi$ **radian**

Q.24 Answer:

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Solution:

$$\begin{aligned}a_n &= a_{n-1}+2n=a_{n-2}+2(n-1+n)\\ &= a_{n-3}+(n-2+n-1+n)\\ &= a_1+2(2+3+\ldots\ldots+n)=n(n+1)\end{aligned}$$

$$\therefore \sum_{i=1}^{20} a_i = 3080$$

Q.25 Answer:

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Solution:

$\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{4}}+\ldots\ldots+\frac{1}{\sqrt{n^2-1}+\sqrt{n^2}}$

Rationalization of D^r

$\therefore \quad S=(\sqrt{2}-\sqrt{1})+(\sqrt{3}-\sqrt{2})+\ldots\ldots+\left(\sqrt{n^2}-\sqrt{n^2-1}\right)$

$\Rightarrow \quad S=n-1$

Q.26 Answer:

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Solution:

$T_m=a+(m-1)d=\frac{1}{n}$

$\Rightarrow \quad md=\frac{1}{n}-(a-d)$ **...(i)**

$T_n=a+(n-1)d=\frac{1}{m}$

$\Rightarrow \quad nd=\frac{1}{m}-(a-d)$ **...(ii)**

(i) $\div$ **(ii)** $\Rightarrow \frac{m}{n}=\frac{\frac{1}{n}-(a-d)}{\frac{1}{m}-(a-d)} \Rightarrow \quad 1-m(a-d)=1-n(a-d)$

$\Rightarrow \quad (a-d)(n-m)=0 \quad \Rightarrow \quad a-d=0 \quad (n\neq m)$

Q.27 Answer:

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Solution:

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We have $t_r = \sum_{k=1}^{r} t_k - \sum_{k=1}^{r-1} t_k$
$= \frac{1}{12} r(r+1)(r+2) - \frac{1}{12}(r-1)(r)(r+1) = \frac{1}{4} r(r+1)$
Now, $\frac{1}{t_r} = \frac{4}{r(r+1)} = 4\left(\frac{1}{r} - \frac{1}{r+1}\right)$
$\Rightarrow \sum_{r=1}^{n} \frac{1}{t_r} = 4 \sum_{r=1}^{n} \left(\frac{1}{r} - \frac{1}{r+1}\right) = 4\left(1 - \frac{1}{n+1}\right) = \frac{4n}{n+1}$

Q.28 Answer:

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Solution:

$T_m = S_m - S_{m-1} = 3m^2 + 5m - 3(m-1)^2 - 5(m-1)$
$= 6m - 3 + 5 = 6m + 2$
Given, $6m + 2 = 164 \Rightarrow m = 27$

Q.29 Answer:

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Solution:

$2^{\frac{1}{4}} \cdot 4^{\frac{1}{8}} \cdot 8^{\frac{1}{16}} \cdot 16^{\frac{1}{32}} \ldots\ldots\ldots$
$= 2^{\frac{1}{4}} \cdot 2^{\frac{2}{8}} \cdot 2^{\frac{3}{16}} \cdot 2^{\frac{4}{32}} \ldots\ldots\ldots\ldots\ldots$
$= 2^{\frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \ldots\ldots\ldots} = 2^S,$
where $S = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \ldots\ldots\ldots$
$\therefore \frac{1}{2} S = \frac{1}{8} + \frac{2}{16} + \frac{3}{32} + \ldots\ldots\ldots$
Subtracting, we get, $\frac{S}{2} = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \ldots\ldots\ldots\infty$
$= \frac{\frac{1}{4}}{1 - \frac{1}{2}} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}, \quad \therefore S = 1$
$\therefore$ **Required sum** $= 2$

Q.30 Answer:

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Solution:

Let there be 2 n terms in the G.P. with first term a and common ratior.

Then,

Sum of all the terms $= 5$ **(Sum of the terms occupying the odd places)**

$\Rightarrow a_1 + a_2 + \ldots + a_{2n} = 5\left(a_1 + a_3 + a_5 + \ldots + a_{2n-1}\right)$
$\Rightarrow a + ar + \ldots + ar^{2n-1} = 5\left(a + ar^2 + \ldots + ar^{2n-2}\right)$
$\Rightarrow a\left\{\frac{1 - r^{2m}}{1 - r}\right\} = 5a\left\{\frac{1 - \left(r^2\right)^n}{1 - r^2}\right\}$
$\Rightarrow 1 + r = 5$
$\Rightarrow r = 4$

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